

Tran Manh Duy

☎ 096 326 9684 | @ tmd.iid2004@gmail.com |  LinkedIn |  GitHub

SUMMARY

Backend software engineer with a strong Computer Science foundation and hands-on experience building scalable, production-ready services. Proficient across multiple languages — Go, Rust, C/C++, TypeScript — and comfortable choosing the right tool for the problem. Experienced designing clean-architecture and hexagonal-architecture backends, implementing domain-driven business logic, and integrating persistence layers (PostgreSQL, MongoDB, Redis). Familiar with authentication patterns (JWT, OAuth2), REST API design with Swagger documentation, and CI/CD pipelines via GitHub Actions and Docker. Eager to contribute to backend systems, API platforms, or data-intensive services at scale.

EDUCATION

University of Engineering and Technology, VNU

BS in Information Technology, CPA: 3.0/4.0

Available for Full-time employment

Hanoi, Vietnam

Sept 2022 – Sept 2026

SKILLS

Programming Languages: Python, Rust, Golang, C/C++, Typescript

Infrastructure: Kafka, Redis, Docker, Git/GitHub, CI/CD (GitHub Actions)

Database: PostgreSQL, MySQL, MongoDB

Language: English, Japanese

EXPERIENCE

Sotatek ., JSC

Hanoi, Vietnam

Rust Developer

Jan 2024 – Present

- Designed and implemented high-performance **RESTful APIs** (using Axum/Actix) and **gRPC services** (using Tonic, Prost) to power CEX/DEX platforms, serving both web and mobile interfaces with low latency.
- Engineered scalable asynchronous messaging systems using **Kafka** and **RabbitMQ** to handle high-frequency trading data and real-time blockchain event processing.
- Optimized database performance and data reliability by integrating **Redis** for caching and **SeaOrm** for type-safe interactions with PostgreSQL.

Blockchain Developer

- Developed and deployed a Starknet Layer 2 solution integrated with Aptos Layer 1, including data availability on Move, Madara chain configuration, and a Rust-based prover for on-chain verification.

Research Assistant

- Researched, documented, and implemented algorithms related to Zero-Knowledge Proofs (ZKPs), including zk-SNARKs and zk-STARKs.

PROJECTS

Kotoba Press Core — Japanese Learning Platform Backend

- Stack:** Go (Gin, Cobra), MongoDB, OAuth2 (JWT), Docker, GitHub Actions, Swagger.
- Designed a **hexagonal architecture** backend (domain / usecase / adapter / port layers) for a Japanese vocabulary and grammar learning platform, keeping business logic fully decoupled from infrastructure.
- Implemented a **Spaced Repetition System (SRS)** using the **SM-2 algorithm** — tracking ease factor, interval, and repetition count per card — with due-date scheduling and per-user deck management persisted in MongoDB.
- Built a **JLPT-aware content API** supporting paginated word browsing, grammar lookup, and polymorphic search across dictionary and grammar collections.
- Developed a **test generation engine** that assembles randomized JLPT mock tests (25 vocabulary + 25 grammar + 3 reading passages) with shuffled answer choices and automated grading.
- Exposed a fully documented **REST API** via Swagger, secured with JWT and OAuth2, with a separate **importer CLI** (Cobra/Viper) for bulk JMdict dictionary ingestion.
- Github:** [Tranduy1dol/kotoba-press-core](https://github.com/Tranduy1dol/kotoba-press-core)

High-Performance E-commerce Microservice

- **Stack:** Rust (Axum, Tokio), PostgreSQL (SeaORM), Redis, Docker, GitHub Actions, k6.
- Architected a scalable backend using **Clean Architecture** and **Cargo Workspace** pattern, decoupling business logic (**core**) from infrastructure (**infra**) with **trait-based port/adaptor** interfaces.
- Solved critical **race conditions** in inventory management ensuring data integrity under high concurrency (verified via Testcontainers integration tests).
- Achieved **~8,000 RPS** with 4.51ms avg latency (P95: 7.61ms) through **Redis caching** strategies and **load testing** (k6).
- Established a **CI/CD pipeline** using GitHub Actions for automated testing and Docker containerization.
- **Github:** [Tranduy1dol/shopping-cart](https://github.com/Tranduy1dol/shopping-cart)